Deep Learning: Group 13 Proposal

Landon Gehr, Julius Pierce, Carson Evans

November 8, 2026

1 Team Roster & Roles

1. Landon Gehr — NEAT implementation — email: lsgehr@mines.edu
2. Julius Pierce — Simulation & Evaluation — email: julius_pierce@mines.edu
3. Carson Evans — Neural Network Implementation — email: carson_evans@mines.edu

2 Project Title & Summary

Comparison of NEAT Network Performance with Standard NNs

We want to explore whether NeuroEvolution of Augmenting Topologies (NEAT), a genetic algorithm that evolves neural network topologies, can produce recurrent networks that compete with standard feed forward neural networks (FFNN) of similar size while adding memory capabilities standard FFNN lack. We would like to assess the performance between the two in both performance and cost. We will examine whether parallel & distributed computing on the HPC can make training practical.

3 Motivation & Relevance

NEAT uses evolutionary algorithms to optimize neural network weights and architectures, while traditional neural networks generally rely on gradient-based optimization. Comparing these approaches can help us understand their relative strengths, particularly for control problems involving temporal dynamics. We are also interested in whether parallel computing can reduce the computational cost associated with evolutionary training. These findings could provide insight into the practicality of neuroevolution for applications such as robotics, autonomous control, and reinforcement learning.

4 Dataset Identification

Since we are working with a simulated physical system, we plan to generate our own data rather than use an existing dataset. One possible environment is Gymnasium's CartPole, which simulates balancing a pole on a moving cart. The simulation provides information such as the cart's position, velocity, and the pole's angle and angular velocity. Our models will use these observations to make control decisions, and their performance will be measured through rewards.

The environment is publicly available at *Gymnasium*[1]. We can generate as much data as needed by running additional simulations. Depending on the models we use, we may need to normalize inputs or organize observations into sequences for recurrent networks.

5 Feasibility & Scope Check

There are existing examples of adding LSTM cells to a NEAT algorithm such as "Evolving Deep LSTM-based Memory Networks using an Information Maximization Objective"[2] and the original NEAT paper [3] itself uses the pole balancing problem as a benchmark.

6 Preliminary Plan

1. **Week 1: Environment Setup (Julius)**
   Select and implement the simulation environment, generate initial data, and establish evaluation metrics.

2. **Week 2: Model Development (Landon and Carson)**
   Implement NEAT with recurrent connections and develop standard feedforward and recurrent neural network baselines.

3. **Week 3: Training and Optimization (All Members)**
   Train models, tune hyperparameters, and explore parallelizing NEAT fitness evaluations using HPC resources.

4. **Week 4: Evaluation and Documentation (All Members)**
   Compare model performance, training time, and network complexity. Analyze results and complete the final report.

7 References

1. `https://gymnasium.farama.org/environments/classic_control/cart_pole/`

2. `https://dl.acm.org/doi/10.1145/2908812.2908941`

3. `https://nn.cs.utexas.edu/?stanley:ec02`